Decentralized HVAC Control System - Visualization Plan

0. Global Design System & Grammar

To ensure the dashboard reads as a single, coherent report, these rules apply strictly across all 5 views.

0.1 Strict Color Palette

- **Zone Identities (Fixed everywhere):**
  - SPACE1-1: Pink-Red (`#ef476f`)
  - SPACE2-1: Yellow (`#ffd166`)
  - SPACE3-1: Teal-Green (`#06d6a0`)
  - SPACE4-1: Blue (`#118ab2`)
  - SPACE5-1: Dark Navy (`#073b4c`)
- **Roles:**
  - Outdoor Air: Warm Orange-Red (`#e76f51`)
  - AHU Supply Air: Cyan-Teal (`#00b4d8`)
  - Electric Energy: Blue (`#3498db`)
  - Gas Energy: Red (`#e74c3c`)
  - Fan Energy: Violet (`#9b59b6`)
  - **Comfort Zone Shading:** Translucent Light Green (`rgba(46, 204, 113, 0.15)`). Used globally across *all* plots to indicate acceptable in-bound ranges (e.g., deadbands, safe CO2 limits).

0.2 Line-Style Grammar

- **Solid Line:** Actual measured/simulated physical values.
- **Dashed Line:** Targets, bounds, or physical limits (e.g., setpoints).
- **Dotted Line:** Internal models or "Ideal Asks".
- **Line Thickness:** Thick (2) for main focus/aggregates. Thin (0.5-1) for baselines, background lines, or individual zone details in aggregate views.

0.3 "Proposed vs Default" Comparison Rule

- **Proposed (MPC):** Full-saturation color, **Solid** line, Width 2.
- **Default (Baseline):** Same hue but 50% desaturated, **Dashed** line, Width 1.5.

0.4 Table Format

All comparison tables must use paired sub-columns: `Metric — Proposed | Metric — Default | Δ` .

View 1: Extended Kalman Filter (EKF) State Estimation Performance

Description: Focuses on a single representative zone to prove the mathematical rigor of the EKF. Highlights unmeasured state inference (Occupancy) and statistical validity.

Plots

1. **Occupancy Estimation vs Ground Truth**
   - **Axes:** X = Time, Y = Number of Occupants.
   - **Data:** True Occupancy (Solid, zone color), Estimated Occupancy (Dotted, zone color).
   - **Styling:** Add a shaded **±2σ uncertainty band** (fill color = zone color, opacity 0.15) behind the estimate line to visually prove the ground truth stays within the filter's confidence intervals.
2. **Residual Zero-Mean Distribution (Violin / KDE Plot)**
   - **Type:** Kernel Density Estimate (KDE) or Violin Plot.
   - **Data:** $y_{T_{in}}$, $y_{W_{in}}$, and $y_{C_{in}}$ plotted side-by-side.

- **Styling:** Proves errors form a bell curve centered exactly at 0 (indicated by a dashed gray reference line).

3. **Normalized Innovation Squared (NIS) & Convergence**
   - **Type:** Split pane time-series.
   - **Top (NIS):** Y-axis uses a **logarithmic scale**. Plot the NIS value with a shaded horizontal gray band from `0.216` to `7.815` (95% chi-square bounds). Proves statistical honesty.
   - **Bottom (Trace P):** Y-axis uses log scale showing the collapse of estimator uncertainty over time.
4. **Final Correlation Matrix (Heatmap)**
   - **Type:** 2D Heatmap of the final state Correlation Matrix (values bounded [-1, 1]).
   - **Axes:** State names on X and Y (T_{in}, W_{in}, N_{occ}, etc.).
   - **Styling:** Diverging colorscale (`RdBu` , centered white at 0). Better than raw covariance as it exposes state confounding without log10 hacks.

Summary Table

- **Contents:** Occ. RMSE, Mean |resid| (T/W/CO2), % NIS in 95% band, Time to 90% convergence.

View 2: Zone MPC vs. Default Controller Performance

Description: Deep dive into **one** representative zone. Compares the MPC against the default EnergyPlus controller.

Plots

1. **Temperature & Comfort Bounds (Time Series)**
   - **Axes:** X = Time, Y = Temperature ($^\circ C$).
   - **Data:** MPC Actual Temp (Solid, full color), Default Temp (Dashed, desaturated), Ideal Ask Temp (Dotted).
   - **Styling:** Standard "Comfort Green" shaded horizontal band representing the allowable deadband.
2. **Relative Humidity Regulation (Time Series)**
   - **Axes:** X = Time, Y = RH (%).
   - **Data:** RH Actuals (Proposed vs Default).
   - **Styling:** Standard "Comfort Green" shaded region between 30% and 60% RH.
3. **Absolute Humidity Request (Time Series)**
   - **Axes:** X = Time, Y = Abs. Humidity Ratio (kg/kg).
   - **Data:** Ideal Ask W (Dotted, zone color), AHU Supply W (Solid, teal).
   - **Styling:** Standard "Comfort Green" shaded region for the absolute humidity equivalent deadband.
4. **Occupancy-Driven CO2 Response (Dual-Axis Time Series)**
   - **Axes:** X = Time, Primary Y = CO2 (ppm), Secondary Y = Occupants (count).
   - **Data:** CO2 (Proposed vs Default) on Primary. True Occupancy on Secondary (thin filled area).
   - **Styling:** Shade the region **above 1000 ppm in translucent red**. The Default controller line should visibly cross into this danger zone, while the Proposed stays below.
5. **Zone Solver Status (Nested Sunburst Pie)**
   - **Type:** 2-Ring Sunburst Chart.
   - **Data:** Outer ring = Stage 1 & Stage 2 Status (Solved, Inaccurate, Failed). Inner ring = Which specific constraint bound the solver (e.g., Temp max, flow min).

Summary Table

- **Columns:** Zone Name | MAE T °C (Proposed | Default) | Max T Error °C (P|D) | % Time T in range (P|D) | MAE RH % (P|D) | % Time RH in range (P|D) | Max CO2 excursion (P|D) | % Time CO2 in range (P|D).

View 3: AHU Coordinator & Multi-Zone Arbitration

Description: Visualizes how the Coordinator handles conflicting requests from all 5 zones simultaneously.

Plots

1. **Temperature Arbitration Envelope (Time Series)**
    - **Axes:** X = Time, Y = Supply Air Temperature ($^\circ C$).
    - **Data:**
        - **Faint Lines:** All 5 Zone Ideal Asks plotted as very thin dotted lines (Width 0.5, Opacity 0.4).
        - **Envelope Fill:** Light shaded region between the max and min ask.
        - **Decision:** The AHU Cooling/Heating Setpoint threading the needle (Solid, thick, Cyan/Red).
2. **Humidity Arbitration Envelope (Time Series)**
    - **Axes:** X = Time, Y = Humidity Ratio (kg/kg).
    - **Data:** 5 faint ideal W requests from zones, the envelope fill, and the final AHU Humidifier/Dehumidifier Setpoint (Solid, Teal).
3. **Ventilation Arbitration Envelope (Time Series)**
    - **Axes:** X = Time, Y = CO2 Request (ppm) or Flow Request.
    - **Data:** 5 faint ideal ventilation requests, the envelope fill, and the final AHU Total OA Flow decision.
4. **AHU Solver Status (Nested Sunburst Pie)**
    - **Type:** 2-Ring Sunburst Chart.
    - **Data:** Outer ring = Solver Status (Solved / Fallback). Inner ring = Active binding constraint (e.g., "CO2 Safety Binding", "Cooling Restriction").

Summary Table

- **Columns:** Metric | Value
- **Rows:** Mean / Max CC setpoint (°C), Mean / Max HC setpoint (°C), % time OA fraction modulating vs floor/ceiling, Solver status breakdown, # times coil bound hit.

View 4: Whole Building Coexistence & Rollup

Description: System-wide macro view proving all zones coexist comfortably on one AHU under the proposed controller, compared to the baseline.

Plots

1. **% Time Within Comfort Band (Grouped Bar Chart)**
    - **Axes:** X = Zones (1-5), Y = % of Simulated Time.
    - **Data:** Grouped bars for T, RH, and CO2. Each variable has a Proposed bar (Solid fill) paired with a Default bar (Desaturated/Hatched fill).
2. **Tracking Error Distribution (Box Plots)**
    - **Axes:** X = Zones, Y = Signed Error (Measured - Setpoint).
    - **Data:** Side-by-side box plots (Proposed vs Default) faceted by T, RH, and CO2. Shows the *severity* of the excursions, not just the percentage of time.
3. **4D Psychrometric Scatter (Side-by-Side Comparison)**
    - **Type:** Two Psychrometric Charts side-by-side (Proposed vs Default).
    - **Axes:** X = Dry-Bulb Temp, Y = Humidity Ratio.
    - **Data:** Scatter trail per zone.
    - **Styling:** Marker Color = Zone Identity. Marker Size = CO2 Concentration. The cluster under "Proposed" will be visibly tighter.
4. **Building-Wide Constraint Heatmap (Time vs. Zone Grid)**
    - **Type:** 2D Heatmap Grid.
    - **Axes:** X = Time of Day, Y = Zone (1 through 5).
    - **Data:** Color intensity showing magnitude of constraint violations.
5. **Zone Compliance (Row of Nested Donuts)**
    - **Type:** 5 nested donut charts arranged horizontally.
    - **Data:** One donut per zone. Inner ring = Temp, Middle = RH, Outer = CO2. Slices colored Green (In range) vs Red

(Out of range).

Summary Table

- **Columns:** Zone Name | % Time T in range (P|D) | % Time RH in range (P|D) | % Time CO2 in range (P|D) | Mean abs. deviation (P|D).
- **Rows:** SPACE1-1 through SPACE5-1, plus a bolded "Building (Mean)" aggregate row.

View 5: Energy Consumption & Efficiency (MPC vs Default)

Description: Translates control smoothness into tangible energy and peak demand savings.

Plots

1. **Instantaneous Power Profile (Stacked Area Chart)**
    - **Axes:** X = Time, Y = Power (kW).
    - **Data:** Stacked areas for Fan Power, Cooling, Heating. Proposed (Solid) vs Default (Dashed outline overlay).
2. **Cumulative Total Energy Use**
    - **Axes:** X = Time, Y = Energy (kWh).
    - **Data:** Proposed (Solid, thick) vs Default (Dashed, thick).
    - **Styling:** Shade the area between the two curves with a translucent fill to highlight the growing energy savings gap.
3. **HVAC Load Duration Curve**
    - **Axes:** X = % of time exceeded, Y = Total HVAC Power (kW).
    - **Data:** Sort instantaneous power descending. Proposed (Solid) vs Default (Dashed).
    - **Rationale:** Proves if the decentralized controller flattens peak electrical demand.
4. **Energy Savings Breakdown (Grouped Bar Chart)**
    - **Axes:** X = Categories, Y = Total Energy (kWh).
    - **Data:** Fan Energy, Cooling Energy, Heating Energy, Total Energy. Grouped by Proposed (Solid) vs Default (Hatched).

Summary Table

- **Contents:** End-uses (Cooling, Heating, Fan, Total), formatted as `Proposed (kWh) | Default (kWh) | Δ (kWh) | Δ (%)`. Includes Peak Power (kW) comparison.